



Crafting Survey Research: A Systematic Process for Conducting Survey Research

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Abstract

Surveys represent flexible and powerful ways for practitioners to gain insights into customers and markets and for researchers to develop, test, and generalize theories. However, conducting effective survey research is challenging. Survey researchers must induce participation by “over-surveyed” respondents, choose

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appropriately from among numerous design alternatives, and need to account for the respondents' complex psychological processes when answering the survey. The aim of this chapter is to guide investigators in effective design of their surveys. We discuss state-of-the-art research findings on measurement biases (i.e., common method bias, key informant bias, social desirability bias, and response patterns) and representation biases (i.e., non-sampling bias and non-response bias) and outline when those biases are likely to occur and how investigators can best avoid them. In addition, we offer a systematic approach for crafting surveys. We discuss key steps and decisions in the survey design process, with a particular focus on standardized questionnaires, and we emphasize how those choices can help alleviate potential biases. Finally, we discuss how investigators can address potential endogeneity concerns in surveys.

Keywords

Survey research · Biases · Survey design · Survey research process · Measurement theory · Common method bias · Key informant bias · Social desirability · Response styles · Non-sampling bias · Non-response bias · Item reversal · Order bias

Introduction: Relevance of Survey Research

Surveys are ubiquitous, used to inform decision makers in every walk of life. Surveys provide practitioners with deeper insights into the attitudes of their customers (e.g., Hohenberg and Taylor (chapter ► [“Measuring Customer Satisfaction and Customer Loyalty”](#)) in this handbook) and employees (e.g., employee satisfaction surveys). Surveys are also helpful in exploring theoretical mechanisms for theory testing and development, as survey research can contribute to generalizing experimental findings to different persons and settings (Krosnick 1999; MacKenzie and Podsakoff 2012). Many relevant and important research questions would be difficult to study without relying on survey data (Hulland et al. 2018). Often, adequate secondary data are not available and experimental manipulations are not feasible. Thus, unsurprisingly, marketing research has a “rich tradition. . . in survey research” (Rindfleisch and Heide 1997, p. 30).

Surveys represent a versatile and powerful research instrument that is applicable in various contexts. For instance, investigators rely on surveys to study:

- Customer attitudes (e.g., customer satisfaction, customer loyalty, voice of the customer surveys)
- Employee attitudes (e.g., employee satisfaction, employee commitment)
- Service quality (e.g., surveys about hotel service)
- Product quality (e.g., surveys with package inserts)
- Performance evaluations (e.g., training evaluation surveys)
- Product feedback (e.g., new product/concept testing surveys)

Recent findings demonstrate the superiority of survey research over other methods. For instance, a recent meta-analysis reveals that direct survey-based techniques more validly indicate consumers’ willingness-to-pay than indirect methods (Schmidt and Bijmolt 2019). Similarly, survey research might deliver the most valid results in studies of sensitive topics (John et al. 2018).

However, despite its important benefits, survey research is in decline (Hulland et al. 2018). Possibly, awareness of potential biases that can occur in survey research may have nurtured skepticism toward surveys, rendering findings less trustworthy or credible. Thus, a critical challenge for survey research lies in separating noise and bias from a survey. As an understanding of how biases emerge will help investigators enhance the validity of their surveys, we discuss the most commonly identified biases in survey research.

Researchers need to make various decisions when developing their surveys. We introduce a systematic process to survey research design that will help investigators organize and structure survey development by answering guiding questions for each stage of the survey research process. In addition, we outline how those decisions can help to alleviate potential biases – an important consideration, as biases from survey research can to a large extent be attributed to “haphazard decisions” (Schwarz 2003, p. 588), investigators make when constructing surveys. While we focus primarily on procedural remedies to avoiding biases (ex ante bias prevention), we also briefly address statistical techniques (ex post bias corrections) and direct readers to further literature. Such statistical techniques represent important supplements to effective survey design.

After reading this chapter, researchers will have an in-depth understanding of the various biases that may affect the results of survey research. In addition, researchers will comprehend the general survey process and know which decisions in survey development will help to reduce potential biases. Figure 1 shows how we have

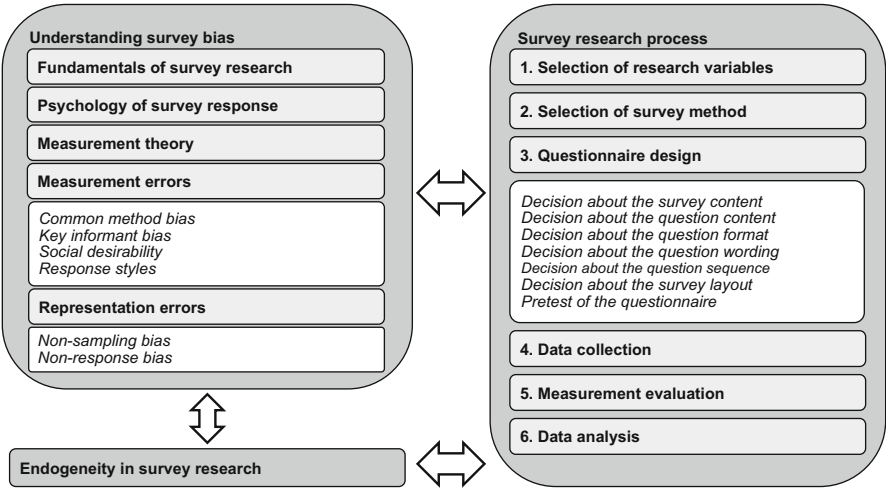


Fig. 1 Overview of chapter structure

structured this chapter. We first foster an understanding of survey bias (section “[Understanding Survey Bias](#)”) by discussing the psychology of survey response (section “[Psychology of Survey Response](#)”) and measurement theory (section “[Measurement Theory](#)”). We then discuss in detail important sources of systematic errors in survey research, which we classify into measurement errors (section “[Sources of Systematic Errors in Survey Research: Measurement Errors](#)”) and representation errors (section “[Sources of Systematic Errors in Survey Research: Representation Errors](#)”). We subsequently outline the survey research process (section “[Survey Research Process](#)”), with a particular focus on how to design the questionnaire (section “[Questionnaire Design](#)”). We briefly address the issue of endogeneity (section “[Endogeneity in Survey Research](#)”) and end by summarizing and aligning the sections (section “[Conclusion](#)”).

Understanding Survey Bias

Fundamentals of Survey Research

A survey comprises a “cross-sectional design in relation to which data are collected predominantly by questionnaire or by structured interview on more than one case (usually quite a lot more than one) and at a single point in time” (Bryman and Bell 2015, p. 63). Surveys can be categorized by several aspects, such as

1. The method in which they are administered to the participant: written, online, telephone, or personal surveys
2. Time horizon: cross-sectional versus longitudinal surveys
3. The role of the respondent: self-reports versus key informants

Survey administration can be classified into personal, telephone, written, and online surveys. We discuss these different forms when outlining the selection of the survey method (section “[Selection of Survey Method](#)”), focusing primarily on written and online surveys because these are the dominant forms of survey research (Hulland et al. 2018).

The time horizon can be purely cross-sectional or longitudinal. While cross-sectional surveys are administered at a single point in time, longitudinal surveys comprise repeated observations for different time periods (e.g., Heide et al. 2007; Jansen et al. 2006; Wathne et al. 2018). Cross-sectional surveys are the dominant form of applied research, encompassing 92.1% (Hulland et al. 2018, p. 94) and 94% (Rindfleisch et al. 2008, p. 262). Examples of longitudinal surveys include

- American Customer Satisfaction Index (theacsi.org; Fornell et al. 1996), which tracks the evolution of customers’ satisfaction with several companies over time
- Harris Poll EquiTrend study (e.g., Vomberg et al. 2015) or Young & Rubicam Brand Asset Valuator (e.g., Mizik and Jacobson 2008), which gauge consumers’ brand perceptions

- The CMO survey, which regularly surveys the opinions of chief marketing officers (e.g., cmosurvey.org)

Longitudinal surveys can be designed in various ways. The American Customer Satisfaction Index and the Harris Poll EquiTrend Study are repeated cross-sectional surveys in which different respondents are sampled each time. Alternatively, researchers can conduct a panel survey in which mostly the same respondents are surveyed each time (chapter ► [“Panel Data Analysis: A Non-technical Introduction for Marketing Researchers”](#) by Vomberg and Wies).

Regarding the role of the respondent, survey participants can either provide self-reports or act as key informants. In self-reports, participants assess questions for themselves. For instance, they may indicate their level of satisfaction or their attitude toward a focal brand. In contrast, key informants provide answers for a higher-order social entity. For instance, employees may indicate the strategic orientation of their company. Key informants are commonly relied on in organizational contexts. We elaborate later on a potential bias stemming from the use of key informants (section [“Key Informant Bias”](#)).

Psychology of Survey Response

Survey researchers need to be aware of the psychological processes that typically occur when participants answer a questionnaire. In business research, participants usually cannot offer predefined responses but form their evaluations when answering the questionnaire. For instance, when customers are asked about their satisfaction with a company, they are not likely to retrieve such an evaluation directly from memory, but instead tend to reflect on their answer when completing the questionnaire. Thus, survey questions trigger a cognitive process of response generation.

It is beyond the scope of this chapter to outline the variety of models that have been proposed to capture these cognitive processes. Therefore, we only briefly summarize the model of Tourangeau et al. (2000), which investigators frequently refer to when studying respondent behaviors (e.g., MacKenzie and Podsakoff 2012; Podsakoff et al. 2003; Weijters et al. 2009).

Tourangeau et al. (2000) argue that respondents pass through five stages when replying to survey questions: (1) comprehension, (2) retrieval, (3) judgment, (4) response selection, and (5) response reporting. In the comprehension stage, participants attend to the survey question and deduce its intent. Respondents then generate a retrieval strategy and search their memories for relevant information. Retrieval thus entails the process of bringing information held in long-term memory to an active state, in which it enters the short-term memory to be used (belief-sampling model). Respondents integrate this information into a judgment (e.g., their satisfaction with a certain product). Finally, when selecting their response, participants map the judgment onto the offered response categories and report their answer (e.g., Krosnick 1999; Tourangeau et al. 2000; Weijters and Baumgartner 2012).

Each of these stages is quite complex and involves a significant amount of cognitive effort by the participant. Thus, during this process, participants may not

be motivated to process survey items in sufficient detail to provide a valid statement. However, even motivated respondents may retrieve biased information. The accessibility-diagnostics theory argues that respondents retrieve information that is accessible to them and has a high diagnosticity (Feldman and Lynch 1988). For instance, very salient but exceptional events (e.g., a specific negative incident with a company) are likely to be more accessible than regular events and thus lead respondents to provide a distorted picture of their true attitudes and opinions. In addition, information provided in earlier questions may represent a source of information that respondents use to form their answers. For instance, the sequence of questions may influence what information respondents retrieve when answering subsequent questions (section “[Decisions About the Question Sequence](#)”). Even if consumers retrieve accurate information, they must make substantial efforts to condense this complex information into rather simple answer categories, such as scales from 1 to 7 (Homburg et al. 2012c).

These examples highlight the complexity of survey response. In the following, we outline the consequences of these psychological processes for the interpretation of survey data.

Measurement Theory

Reliability and Validity: Fundamentals

Measurement theory claims that any observed value x for a question (e.g., an observed value for the liking of a brand) is the sum of a true value t (also referred to as trait) and measurement error, which can have a random e and a systematic component s . Hence, any observed value can be understood in the following way (Eq. 1):

$$x = t + s + e \quad (1)$$

Importantly, the random error component poses threats to the reliability of a survey question, and the systematic error component can affect the question’s validity. A survey question can be considered reliable when it produces the same results under the same measurement conditions, whereas the question has validity when it actually measures what it purports to measure. An intuitive example to understand the concept of random and systematic error is the following. Imagine that 100 researchers measure the time it takes for a participant to run a certain distance. Usually, when all researchers compare their results, their observed measurements will differ slightly. Thus, individual measurements likely suffer from random measurement error. However, since this error is assumed to be randomly distributed among participants, its expected value is zero: $E(e) = 0$. Hence, when taking the average value, researchers likely obtain an unbiased measure (Iacobucci 2013).

With respect to survey research, a characteristic such as imprecise wording can raise fundamental threats to reliability: participants could interpret words such as “usually” or “almost” differently, adding noise to the data

(section “[Decisions About the Question Wording](#)”). However, since the expected value of the random error is zero, survey researchers can ask multiple questions when measuring abstract concepts such as commitment or satisfaction. Just as in the stop watch example, averaging multiple measurements may help to safeguard against reliability concerns (section “[Decisions About the Question Content](#)”).

In contrast, systematic errors affect the validity of survey questions. Validity refers to the degree to which a measure really measures what it is supposed to measure. Since the expected value of the systematic error is not zero ($E(s) \neq 0$), repeated measurements cannot alleviate potential validity concerns. Intuitively, this can be explained by continuing with the stop watch example. If all researchers had received stop watches that systematically add 10 s, even the average of the individual measurements will be biased. We discuss sources of systematic errors in survey research in sections “[Sources of Systematic Errors in Survey Research: Measurement Errors](#)” and “[Sources of Systematic Errors in Survey Research: Representation Errors](#).”

Reliability and Validity: Implications for Survey Research

A natural follow-up question is the extent to which random and systematic errors influence the results of survey research. Many times survey researchers are interested in establishing relationships between variables. In the simplest case, investigators can focus on bivariate correlation coefficients, and in the following we discuss the bivariate correlation coefficient between a variable x (e.g., customer satisfaction (CS)) and a variable y (e.g., word-of-mouth behavior (WOM)). In line with Eq. 1, we assume that both variables are measured with error. We apply two common assumptions in deriving the correlation coefficient: we assume (1) uncorrelated random measurement errors (i.e., $\text{Cov}(e, t) = 0$; $\text{Cov}(e, s) = 0$; $\text{Cov}(e_x, e_y) = 0$) and (2) no correlation between the true value and the systematic measurement error ($\text{Cov}(t, s) = 0$). These assumptions lead to the following correlation coefficient (Eq. 2) (e.g., Baumgartner and Steenkamp 2001; Homburg and Klarmann 2009):

$$\begin{aligned}
 r(x, y) &= \frac{\text{Cov}(CS; WOM)}{\sqrt{\text{Var}(CS) \cdot \text{Var}(WOM)}} = \frac{\text{Cov}(x; y)}{\sqrt{\text{Var}(x) \cdot \text{Var}(y)}} = \\
 &= \frac{\text{Cov}(t_x + e_x + s_x; t_y + e_y + s_y)}{\sqrt{\text{Var}(t_x + e_x + s_x) \cdot \text{Var}(t_y + e_y + s_y)}} = \quad (2) \\
 &= \frac{\text{Cov}(t_x, t_y) + \text{Cov}(s_x, s_y)}{\sqrt{[\text{Var}(t_x) + \text{Var}(e_x) + \text{Var}(s_x)] \cdot [\text{Var}(t_y) + \text{Var}(e_y) + \text{Var}(s_y)]}}.
 \end{aligned}$$

Equation 2 offers three important insights. First, a common concern regarding survey research is that participants may provide inflated answers that bias the results (e.g., De Jong et al. 2015). For instance, managers may be tempted to exaggerate their performance. However, while these over- or understatements affect mean

values, mean values do not directly affect Eq. 2. Thus, (systematic) over- or understatements do not bias the relationships between variables in survey research.

Second, in many research applications, investigators are interested in the direction of a relationship rather than in the size of the coefficient. Since the denominator in Eq. 2 contains only variances that cannot become negative, only the numerator is responsible for the direction of the correlation coefficient. Increased variances in the denominator can only reduce the size of the correlation coefficient. Consequently, Eq. 2 reveals that random measurement errors cannot change the sign of the correlation coefficient since the random error is only part of the denominator. From the perspective of a survey researcher, Eq. 2 implies that random measurement errors can lead to only conservative results by decreasing statistical power. Thus, random measurement errors may obscure an effect that is present, leading to Type II errors. However, random measurement errors cannot artificially create relationships.

Third, the impact of systematic measurement errors (also referred to as method error) can be twofold since the systematic error is part of both the numerator and the denominator in Eq. 2. If sources of systematic errors affect both variables independently (i.e., $\text{cov}(s_x, s_y) = 0$), then systematic errors have the same impact as random errors: they can lower statistical power but cannot artificially create effects. However, more likely sources of systematic errors (e.g., key informant bias) affect both variables simultaneously (i.e., $\text{cov}(s_x, s_y) \neq 0$). Thus, systematic errors can affect not only the strength but also the direction of an effect. Systematic errors could be responsible for either detecting artificial relationships in cases in which there is no true relationship (i.e., Type I error) or masking existing relationships (i.e., Type II error) (Baumgartner and Steenkamp 2001; Homburg et al. 2012c). Huang and Sudhir (2021) provide compelling empirical evidence for the latter. They show that systematic biases in survey research can lead to conservative estimates, that is, an underestimation of true effect sizes.

Sources of Systematic Errors in Survey Research: Measurement Errors

In this section, we review key sources of systematic biases that may affect the validity of survey research. A bias is “any force, tendency, or procedural error in the collection, analysis, or interpretation of data which provides distortion” (Crisp 1957, cited in Tortolani 1965, p. 51). Knowledge regarding these threats is important because, to a certain extent, researchers can safeguard against them when designing the survey instrument.

In the following, we describe the most commonly discussed biases in survey research. Following prior investigators, we categorize those biases into measurement errors and representation errors (e.g., Baumgartner and Steenkamp 2006). Measurement errors reflect tendencies of respondents to answer to survey questions on some grounds other than the item content. Specifically, we discuss common method bias (section “[Common Method Bias](#)”), key informant bias (section “[Key Informant Bias](#)”), social desirability bias (section “[Social Desirability](#)”), and response patterns

(section “[Response Styles](#)”). Representation errors reflect biases due to the selection of the sample of respondents. These biases could follow from unrepresentative sampling frames – non-sampling bias (section “[Non-sampling Bias](#)”) – or participants’ unwillingness or inability to respond – unit non-response bias (section “[Non-response Bias](#)”).

Common Method Bias

Conceptualization. Common method bias (CMB) – one of the most frequently discussed threats to survey research – can largely undermine the validity of a study’s findings (Hulland 2019; Palmatier 2016). However, no agreement on a definition of CMB presently exists. CMB definitions differ widely in scope. Podsakoff et al. (2003, p. 880) favor a broad definition and refer to CMB as “variance that is attributable to the measurement rather than the construct of interest.” Under this definition, CMB can be regarded as an umbrella term for various biases that can be classified into four categories (Podsakoff et al. 2003, p. 882):

- *Common source rater:* for example, consistency motif, social desirability (Section “[Social Desirability](#)”), response patterns (section “[Response Styles](#)”), or a tendency toward satisficing
- *Item characteristics:* for example, item ambiguity (section “[Decisions About the Question Wording](#)”), common scale formats or anchors (section “[Decision About the Question Format](#)”)
- *Effects due to item context:* for example, item embeddedness (section “[Decisions About the Question Sequence](#)”)
- *Measurement context:* for example, independent and dependent variables measured at the same point in time

A narrow definition of CMB focuses on the last category and attributes the distortion of the sample’s covariance structure to the use of the same data source to measure both independent and dependent variables (e.g., Klarmann 2008; Rindfleisch et al. 2008). We adopt this definition and discuss other biases that Podsakoff et al. (2003) mention separately.

CMB can emerge when a single informant provides information on the independent and dependent variables. As we elaborate below, the severity of CMB will then depend on various factors, such as the type of collected measure (lower CMB threat for objectively verifiable measures), the complexity of the tested relationships (lower CMB threat for quadratic and interactive relationships), whether a time lag occurs in data collection (lower CMB threat when independent and dependent variables are collected at different points in time), or the response format (lower CMB threat if the independent and dependent variables are measured in different scale formats).

CMB can undermine the validity of survey research in two ways. First, it can lead to biases in estimates of construct reliability and validity, as it can inflate reliability and validity measures by 18 to 32% (Podsakoff et al. 2012, p. 543). Thus, CMB can lead researchers to mistakenly believe that they have validly measured the constructs of interest when in reality they have captured method artifacts.

Second, CMB can bias the parameter estimates between two constructs. Notably, however, investigators document vastly different results when quantifying the impact of CMB: some evidence indicates that, on average, CMB inflates correlation coefficients between constructs by approximately 45%, with a range of 27% to 304% (MacKenzie and Podsakoff 2012, p. 543; Podsakoff et al. 2012, p. 545). However, a meta-analysis of 42,934 correlation coefficients showed correlations based on single-source self-reports to be on average only 0.02 higher than correlation coefficients from different sources (Crampton and Wagner 1994).

Reasons for occurrence. To effectively address CMB, understanding the reasons for its occurrence is important. Given the pivotal role CMB assumes in marketing research projects, the literature not surprisingly offers plenty of explanations for why CMB may occur (Podsakoff et al. 2003 provide a detailed discussion). In the following, we highlight selected explanations.

First, when reading the questions, respondents may start developing implicit theories of the relationships between the constructs. In effect, their answers then represent reflections of these theories. For instance, managers might be asked to evaluate the innovativeness of their business unit and their competitive position. If the participating managers develop the implicit theory that innovativeness represents a focal driver of performance, then their answers referring to the business unit's innovativeness might be influenced by prior answers on the unit's competitive position (e.g., Podsakoff et al. 2003).

Second, respondents strive for consistency in their cognitions. Therefore, they may try to provide answers that they think are consistent. They may search for similarities in the questions and adapt their answers accordingly. As a consequence, stated answers may not truly reflect their behaviors or opinions.

While these two explanations in general outline why CMB can emerge, an important question is, in which situation is CMB more likely to occur? Empirical evidence suggests that two factors drive potential CMB concerns: (1) the nature of the constructs under investigation and (2) the complexity of the investigated relationships.

The literature rather consistently states that the likelihood of CMB to occur depends on the nature of the questions. For example, the percentage of common method variance is lower in marketing (16%) than in psychology or sociology (35%) (Cote and Buckley 1987). Constructs investigated in psychology or sociology are likely to be abstract and complex and thus be harder to answer and may trigger cognitive processes that increase covariation between systematic error components (MacKenzie and Podsakoff 2012; Podsakoff et al. 2012; section “[Psychology of Survey Response](#)”). Empirical evidence confirms these expectations. In general, concrete, externally oriented, and verifiable constructs are less prone to CMB than are abstract, internally oriented, and non-verifiable constructs (Chang et al. 2010; Rindfleisch et al. 2008).

Second, with increasing complexity of the investigated relationships, CMB is less likely to occur. Analytical evidence demonstrates that CMB can deflate quadratic and moderating relationships but cannot create them (Siemens et al. 2010). For such complex relationships, participants are unlikely to develop corresponding implicit

theories that affect their responses. Therefore, if the main interest of research lies in identifying quadratic relationships or interaction effects, CMB is not likely to undermine researchers' findings (Vomberg et al. 2020).

Procedural remedies. A general recommendation is that investigators should favor procedural remedies over statistical remedies when addressing potential biases in survey research. While statistical controls are rather symptom-based – that is, they target the consequences of CMB only in the analysis stages – procedural remedies try to eliminate sources of CMB in the moment of collecting the data. Naturally, in many situations researchers will not be able to completely alleviate CMB concerns with procedural remedies, and in these cases, statistical remedies can help to elevate the credibility of the findings. In the following, we discuss four procedural remedies: (1) use of different data sources, (2) temporal separation, (3) proximal separation, and (4) psychological separation of scale of scale formats.

(1) *Different data sources.* An investigator nullifies the risk of CMB when relying on different data sources for the independent and dependent variables (Rindfleisch et al. 2008, p. 274; Ostroff et al. 2002). This approach makes it impossible for participants to develop implicit theories between independent and dependent variables.

First, researchers can survey different respondents to evaluate the independent and dependent variables (Gruner et al. 2019) – that is, use dyadic data: one portion of the sample is used to estimate the independent variables and the remaining portion evaluates the dependent variables. Empirical evidence supports the effectiveness of dyadic data to attenuate CMB. Correlations of independent and dependent variables when rated by one respondent ($r = 0.359$) dropped by 49% when different respondents evaluated them ($r = 0.184$) (Podsakoff et al. 2012).

Although this procedure appears promising, it requires large sample sizes, and therefore is not appropriate for all kinds of surveys. Particularly in organizational research, researchers observe generally declining response rates, making it especially challenging to recruit additional respondents (section “Non-response Bias”). It might also be problematic in small companies where the owner is in charge of most of the decisions (Rindfleisch et al. 2008).

Second, researchers can rely on a combination of secondary data and survey data. Most commonly, researchers collect independent variables via a survey study and evaluate performance outcomes from profit and loss statements (e.g., Vomberg et al. 2020). Research suggests that this approach is also effective. Obtaining independent and dependent variables from different data sources can reduce their correlations by 49% (Podsakoff et al. 2012, p. 548).

However, relying on different sources might not be feasible in several settings. For instance, use of different data sources is not viable when the two variables cannot be validly inferred from different sources (e.g., self-referential attitudes and perception constructs; Johnson et al. 2011), when archival data cannot adequately represent the construct, or when such data are available only for prohibitively high costs in terms of money or time.

(2) *Temporal separation of measurement.* An alternative might be to separate the collection of independent and dependent variables temporally by including a

time lag (e.g., Homburg et al. 2020; Jansen et al. 2005; Vomberg et al. 2020). Empirical research indicates that temporal separation of measurement can be effective for reducing potential CMB. However, effectiveness depends on the length of the time lag, with longer time lags triggering the possibility that intervening events might introduce new sources of biases (Podsakoff and Organ 1986; Chang et al. 2010). While some research indicates that temporal separation is effective for same point in time versus two weeks later (Johnson et al. 2011) and same point in time versus one month later (Ostroff et al. 2002), other work finds no significant improvement with relatively long time lags (30 vs. 36 months) (Rindfleisch et al. 2008).

(3) *Proximal separation*. Researchers can increase the proximal distance between independent and dependent variables in the questionnaire. As we elaborate in section “[Decisions About the Question Sequence](#),” separating measures significantly reduces shared variance (Weijters et al. 2009). For instance, concern regarding CMB can be alleviated by first measuring the dependent and then the independent variables.

(4) *Psychological separation of scale formats*. Researchers frequently rely on common scale formats for a variety of questions within a questionnaire. While common scale formats make answering the survey questions easier, they enhance the probability that cognitions formed when answering one question may be retrieved when answering another question. However, relying on different scale formats when assessing the independent and dependent variables reduces potential CMB (e.g., Vomberg et al. 2020), as different formats disrupt potential consistency bias (Feldman and Lynch 1988; Podsakoff et al. 2012; Rindfleisch et al. 2008).

Initial empirical evidence supports the effectiveness of using different scale formats. For instance, labeling anchor points differently for independent and dependent variables shrinks CMB by 18% (Johnson et al. 2011). Even larger decreases of 38% or 60% occur when the scale format is changed completely (Arora 1982; Kothandapani 1971).

Statistical remedies. The literature contains discussions of several statistical remedies for reducing CMB. Since these remedies are beyond the scope of the article, we briefly mention only a few. Podsakoff, MacKenzie, and Podsakoff (2012) provide a detailed overview of several statistical techniques, and Hulland et al. (2018) outline how to test for CMB. Hulland et al. (2018) discuss the correction-based marker variable technique, which is one of the most frequently applied approaches to address CMB in marketing research. This approach suggests that a marker variable that is theoretically unrelated to the constructs under investigation resembles the amount of CMB (Lindell and Whitney 2001). Grayson (2007, Appendix B) intuitively summarizes this approach.

Other approaches include (1) the measured latent factor technique, or measured response style technique, in which the researcher directly measure sources of CMB (e.g., Wathne et al. 2018), (2) the unmeasured latent factor technique, in which researchers model a method factor in structural equation models (e.g., Homburg et al. 2011), and (3) endogeneity-correction approaches, in which researchers employ instrument-free techniques (e.g., Vomberg et al. 2020).

Key Informant Bias

Conceptualization. Survey participants can act either as respondents providing self-reports or as key informants. While respondents describe their own beliefs, attitudes, opinions, or feelings, key informants generalize about patterns for a higher social unit (e.g., a company's culture) on the basis of their experiences (Seidler 1974). For example, in an employee satisfaction survey, employees act as respondents when they evaluate their own job satisfaction level. However, they act as key informants when they assess an organizational culture (higher social entity). Key informant bias can occur in the latter case; it comprises the distortion of the sample's covariance structure that arises because the data collection has taken place through key informants. As we elaborate below, the severity of key informant bias will depend on various factors, most pressingly on the expertise of the key informant (less bias if the key informant is knowledgeable, has a higher hierarchical position, and has longer job tenure) and the type of collected data (less bias for objectively verifiable measures).

To the best of our knowledge, only two studies systematically evaluate the occurrence of key informant bias (Homburg et al. 2012b; Kumar et al. 1993), and the two studies deliver conflicting conclusions. Kumar, Stern and Anderson (1993, p. 1646) report high levels of key informant *disagreement*, whereas Homburg et al. (2012c, p. 605) find in their meta-analytical study high levels of key informant *agreement* (0.872).

In addition, Homburg et al. (2012c) investigate how large the systematic error provoked by key informant bias needs to be to create artificial effects or to mask true effects. They demonstrate that the requisite size depends largely on the study's sample size and the strength of the effect, and conclude that in the studies of their meta-analysis there is a risk that small and medium-sized effects were not correctly identified.

Reasons for occurrence. Key informant bias can emerge if investigators have unrealistic expectations about the key informant's knowledge. For instance, it might not be feasible for key informants to provide complex judgment about organizational characteristics that may result in random measurement error. In addition, key informant bias may emerge owing to positional bias or knowledge deficiencies (Phillips 1981). For instance, key informants' backgrounds can substantially influence the answers they provide (Homburg et al. 2005). When marketing and sales managers were asked to evaluate the marketing function (same higher social entity), mean values differed considerably between key informants from the two departments (Fig. 2). Sales managers perceive marketing managers (on a scale from 0 to 100) to have a rather low level of customer know-how (mean value = 37), whereas marketing managers on average consider themselves to have rather high levels customer know-how (mean value = 59).

In addition, in line with empirical findings on CMB, the risk of key informant bias largely hinges on the nature of the studied constructs. Key informant reports are reliable for objective and salient issues from the present (e.g., performance outcomes). However, for more abstract measures (e.g., organizational culture) key informants tend to be less accurate. Furthermore, a higher hierarchical position and tenure also increase reliability (Homburg et al. 2012c; Kumar et al. 1993).

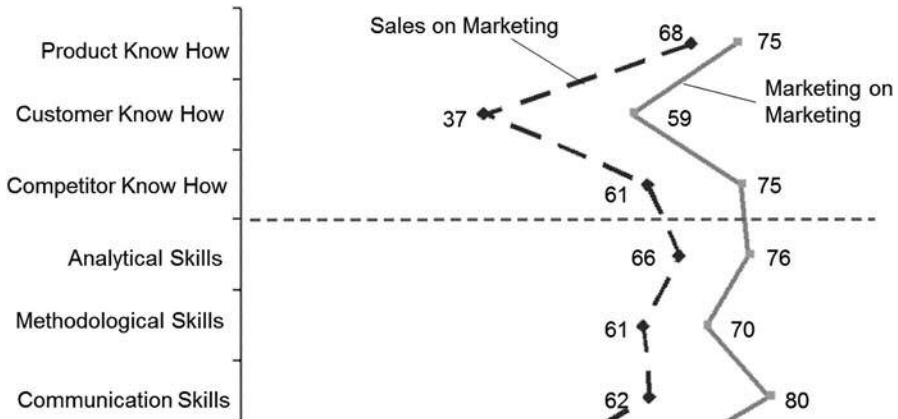


Fig. 2 Marketing and sales managers' evaluations of the marketing function – comparison of mean values (Homburg et al. 2005, p. 14, reproduced with the kind permission of the publisher)

Procedural remedies. The most important and effective remedy to alleviate concerns regarding key informant bias is the careful selection of key informants. The investigator needs to carefully align the contacted key informants with the objective of the study. Typically, key informants are not chosen randomly and are not deemed representative for the population. Instead, researchers select respondents who have special qualifications, such as their position in the company or their knowledge (Phillips 1981; Kumar et al. 1993).

The question of whom to contact as key informants depends primarily on the research context. For instance, if the research objective focuses on strategic issues, high-ranked key informants are probably appropriate. However, if the study investigates operational aspects, such as specific sales approaches used, then key informants with regular customer contact are more promising.

In addition, questions that require less demanding social judgments and instead are impersonal and focus on objective and observable phenomena seem preferable (Phillips 1981; Homburg et al. 2012c; Klarmann 2008). However, whether this condition is realistic depends largely on the context of the study. Nevertheless, even if more complex or abstract constructs are the focus of the study, the researcher might lower the risk of key informant bias with careful wording of the questionnaire items (section “[Decisions About the Question Wording](#)”).

Finally, if key informant response accuracy is expected to be low or empirical evidence for reliability is scarce in the particular research domain, the researcher may use triangulation – that is, combining methods in the study of the phenomenon. For instance, investigators can try to survey multiple key informants per company. However, since triangulation is costly, sometimes difficult to implement (Rindfleisch et al. 2008), and devoid of substantial additional value if the responses of the first respondent have been accurate (Homburg et al. 2012c), this option should be carefully chosen.

Statistical remedies. Statistical procedures address either the reliability or validity of the key informant study. They aim either to establish or enhance reliability or validity (Homburg et al. 2012c) provide a systematic overview of various approaches). For instance, authors can establish reliability by demonstrating key informants' job tenure or experience with the company (Kumar et al. 1993) or by measuring their perceived knowledge level (e.g., Ghosh and John 2005; Kumar et al. 2011). In the case of triangulated data, researchers can also employ correlational approaches, such as intraclass correlation ICC(1) (Bliese 2000; chapter ► “Multi-level Modeling” Haumann et al. in this handbook) or the absolute deviation agreement index (Burke and Dunlap 2002; LeBreton and Senter 2008). To enhance validity, researchers can integrate factors representing the data sources into a structural equation model, allowing estimation of trait relationships while controlling for systematic error (e.g., Cote and Buckley 1987).

Social Desirability

Conceptualization. As one researcher noted over 20 years ago, “One well-known phenomenon in survey research is overreporting of admirable attitudes and behaviors and underreporting of those that are not socially respected” (Krosnick 1999, p. 545). This phenomenon is called social desirability and represents the distortion of the sample's covariance structure that arises owing to the tendency of respondents to reply in a manner that others will view favorably (e.g., De Jong et al. 2010). For instance, in a study on consumer innovativeness, 41% of the respondents indicated they owned, had repurchased, or had seen products that did not actually exist in the market (Tellis and Chandrasekaran 2010).

While social desirability is often considered a source of bias in survey research (e.g., Podsakoff et al. 2003), surprisingly few investigators explicitly address it in their studies (Steenkamp et al. (2010) and Tellis and Chandrasekaran (2010) represent some notable exceptions). As a consequence, knowledge about the impact of social desirability on survey results is scarce.

Reasons for occurrence. Theory proposes two factors to explain the occurrence of socially desirable responding: the level of awareness (self-deception vs. impression management) and the domain of content (agency-based vs. communion-based contexts). Self-deception occurs when participants *unconsciously* dissemble, and impression management occurs when participants *consciously* dissemble (e.g., Krosnick 1999; Paulhus 1984).

Regarding the domain of content, some respondents are more likely to engage in socially desirable responding because they have a need to be perceived as more powerful than others (egoistic response tendencies, arising in agency-based settings). Other respondents offer socially desirable responses because they have a need to be perceived as exceptionally good members of society (moralistic response tendencies, emerging in communion-based contexts) (Paulhus and John 1998; Paulhus 2002; Steenkamp et al. 2010).

Procedural remedies. Self-deception and impression management are particularly important in controlling or reducing potential social desirability bias. Socially desirable responding owing to self-deception might be part of a respondent's

personality, and therefore should not be controlled for to prevent elimination of a central component of individual differences in personality. However, the impression management component should be controlled for because this component embodies a conscious bias (Paulhus 1984).

Several methods are available to prevent or at least reduce socially desirable responses to questionnaires. Survey researchers should assure respondents that their answers are anonymous, reinforce that the items have no right or wrong answers, emphasize that people hold different opinions about the issues addressed in the questionnaire, and encourage respondents to answer as honestly as possible. Paradoxically, socially desirable responding remains an issue under anonymous conditions when there is no one to impress (Mick 1996).

Another remedy is to allow respondents to report on the external world rather than answering questions about themselves. The solution is to ask indirect questions in the neutral third-person form that are not affected by the social desirability bias. This approach is based on the assumption that respondents project their opinion onto others and consequently give more honest answers, an assumption that Fisher (1993) demonstrates empirically. In addition, the social distance between interviewer and respondent should be reduced (Nederhof 1985), and socially sensitive questions should be eliminated or placed at the end of the questionnaire to avoid carry-over effects (Baumgartner and Steenkamp 2006).

Finally, a third way to deal with a social desirability bias is through randomized response techniques (e.g., De Jong et al. 2010, 2015; Himmelfarb and Lickteig 1982; Warner 1965). In this approach, respondents are asked a sensitive question (e.g., “Are you willing to pay higher prices for sustainable products?”) with response options of “yes” and “no.” Prior to answering the question, they flip a coin and adapt their answers following the outcome of the coin flip. If the coin flip returns “heads,” respondents should answer the question with “no” regardless of whether they truly engaged in the questioned behavior. However, if the flip returns “tails,” respondents should answer the question truthfully with “yes” or “no.” Since investigators cannot see the outcome of the coin flip, they cannot tell whether a particular “no” response denotes a negative answer that reveals the respondent’s true attitude or a coin flip that has come up heads (or both). Note that in this simplistic illustration, only the socially less desirable answer (i.e., “no”) is concealed (De Jong et al. (2010) describe more advanced designs). In theory, this technique should counteract a social desirability bias. However, initial empirical evidence indicates that randomized response techniques can also deliver worse results than directly asking participants (Holbrook and Krosnick 2010; John et al. 2018).

Statistical remedies. The literature offers two main approaches to check or control for social desirability: (1) including a measured latent factor capturing social desirability in the empirical model (e.g., Podsakoff et al. 2003) or (2) correlating separate survey measures with a measured social desirability score (Steenkamp et al. (2010) develop a systematic process to check for social desirability concerns). The latter approach demonstrates whether a social desirability bias likely affects the constructs in the study and the first approach is intended to control for social desirability in empirical models. However, both approaches require the investigator to measure the

social desirability construct. To do so, literature suggests relying on Paulhus's (2002) *Balanced Inventory of Desirable Responding* (e.g., Steenkamp et al. 2010).

Response Styles

Response styles lead to a distortion of the sample's covariance structure that arises because, regardless of the question content, respondents favor certain response categories (e.g., Van Rosmalen et al. 2010). The most frequently discussed response styles are respondents' tendency to select specific subsets of response options such as disproportionately favoring the positive side of a response scale (acquiescence response style, or yay-saying).

Response styles can compromise the comparability of the data: the same responses can have different meanings for different respondents. Participants may display different response styles between countries (e.g., Tellis and Chandrasekaran 2010) or between different modes of data collection. Response style biases are higher in telephone interviews than in written and online surveys (Weijters et al. 2008). Response styles thus undermine the ability to validly compare mean values. Furthermore, response styles can also influence construct variances and correlations between constructs. For instance, researchers found that response styles account for 8% of construct variance in their sample and for 27% of the variance in the observed correlations (Baumgartner and Steenkamp 2001).

However, despite their potentially biasing effects, response styles have not received much attention in the marketing literature. In the following, we discuss two common groupings of response styles: (1) acquiescence, disacquiescence, and net acquiescence and (2) extreme responding, midpoint responding, and response range.

Acquiescence, disacquiescence, and net acquiescence. Acquiescence, also called yay-saying, is the respondent's tendency to agree with items regardless of their content, whereas disacquiescence, or nay-saying, is the respondent's tendency to disagree with items regardless of their question content (Baumgartner and Steenkamp 2001; Tellis and Chandrasekaran 2010). Net acquiescence is acquiescence minus disacquiescence and reflects the tendency to show greater yay- than nay-saying (Baumgartner and Steenkamp 2001; Tellis and Chandrasekaran 2010). However, many researchers do not distinguish between acquiescence and net acquiescence (e.g., Greenleaf (1992) uses the label acquiescence but actually measures net acquiescence). We focus on acquiescence and disacquiescence.

Acquiescence poses problems for segmentation research because it can lead to the emergence of clusters that reflect response styles rather than attitudinal information (Greenleaf 1992). In addition, it may falsely heighten correlations among items that are worded in the same direction (Winkler et al. 1982). Remarkably, although acquiescence and disacquiescence appear to be opposites, empirical evidence demonstrates only small- to medium-sized negative correlations between them (Baumgartner and Steenkamp 2001: $r = -0.16$; Tellis and Chandrasekaran 2010: $r = -0.31$).

Dispositional (e.g., personality traits) and situational factors (e.g., item ambiguity) may explain why the three response styles occur (Knowles and Condon 1999).

Regarding dispositional factors, research has demonstrated inconsistent results for demographic variables (particularly gender). For instance, research shows that respondents with an acquiescence response style tend to be extroverted, impulsive, emotional, and undercontrolled. Respondents prone to a disacquiescence response style are likely to be introverted, cautious, rational, and overcontrolled. Acquiescence may be evoked by a desire to please, to be agreeable in social situations, or to display deference to the researcher (e.g., Krosnick 1999).

However, findings on the participants' cultural backgrounds are more consistent. Respondents from countries scoring high on collectivism (uncertainty avoidance) tend to display more (less) acquiescence in their response style (Tellis and Chandrasekaran 2010).

Regarding situational factors, participants are more likely to display an acquiescence response style if response categories are fully labeled (Weijters et al. 2010a), if response categories contain a neutral point (Weijters et al. 2010a), or if items are ambiguous (Podsakoff et al. 2003). Similarly, the personal situation of the participant can provoke an acquiescence response style. An acquiescence response style is likely to occur if participants read items uncritically (Messick 2012), if they experience time pressure (Baumgartner and Steenkamp 2006), or if their cognitive capabilities are exceeded (e.g., items at the end of a long questionnaire) (MacKenzie and Podsakoff 2012).

Extreme responding, midpoint responding, and response range. A respondent with an extreme response style tends to favor the most extreme response categories regardless of the item content. Midpoint responding refers to the tendency to use the middle-scale category regardless of the question content. Response range refers to the tendency to use a wide or narrow range of response categories around the mean response (Baumgartner and Steenkamp 2001).

Thus far, few conceptual and empirical investigations have focused on midpoint responding and response range. Current knowledge suggests that response ranges (and also an extreme response style) may relate to the characteristics of the respondent; they concern rigidity, intolerance of ambiguity, and dogmatism and are associated with higher levels of anxiety and possibly deviant behavior (Hamilton 1968).

In the last few years, investigators have become increasingly interested in extreme responding, with research linking it to personality (Cabooter et al. 2012; Naemi et al. 2009), scale format (Weijters et al. 2020), language (De Langhe et al. 2011; Weijters et al. 2013), and culture (De Jong et al. 2008) – although these papers do not necessarily use the term extreme response style (e.g., De Langhe et al. (2011) talk about “anchor contraction”). In addition, a recent literature stream on item response trees (IRTtree) models investigated extreme (and midpoint) responding (e.g., Böckenholt 2012, 2017; Zettler et al. 2015).

Procedural remedies. Thus far, few procedural remedies have been identified for addressing biases from response styles. To the best of our knowledge, the literature provides only initial remedies for midpoint responding and for an acquiescence and a disacquiescence response style.

First, to address midpoint responding, some authors suggest eliminating the middle response category or including a “don't know” category (Baumgartner and

Steenkamp 2006; Schuman and Presser 1996). Second, since acquiescence is more likely for positively worded items and disacquiescence for negatively worded items (Baumgartner and Steenkamp 2001), investigators suggest introducing doubly balanced scales (Tellis and Chandrasekaran 2010). However, researchers need to carefully evaluate this option since negatively worded items may lead to misresponse (Weijters et al. 2010b; section “[Decisions About the Question Wording](#)”).

Finally, response patterns are often problematic in international market research. Our recommendation is that researchers should make sure that all respondents receive a similar response format with equally familiar labels, preferably in their native language.

Statistical remedies. To investigate whether response styles likely bias the survey results, investigators should correlate the focal constructs of interest with measures of the different response styles – that is, use a measured response style technique. If this analysis does not demonstrate potential biases, then the investigators should proceed as planned. However, if potential threats to validity emerge, the authors should include measures of different response styles in their models (e.g., Weijters et al. 2008).

The literature proposes various ways in which response styles can be measured. For example, response styles can be calculated on the basis of items that are included in the survey – it is not necessary to add items to specifically measure response styles (Weijters et al. 2008, 2010b; Tellis and Chandrasekaran 2010). Alternatively, a latent-class model can identify different response styles (Rosmalen et al. 2010).

Sources of Systematic Errors in Survey Research: Representation Errors

Non-sampling Bias

Conceptualization. Investigators in many research projects are interested in generalizing findings to an overall population. Therefore, representativeness of the sample is important. That is, the sample needs to display approximately the same characteristics as the overarching population. However, as Fig. 3 illustrates, two biases may threaten representativeness: non-sampling bias and nonresponse bias. Non-sampling bias refers to the distortion of the sample’s covariance structure that arises because the population is not adequately represented in the original sample (i.e., the sample that received the questionnaire does not resemble the overall population). Non-response bias is a distortion of the sample’s covariance structure that arises because the structure of the final sample does not correspond with the structure of the original sample. We discuss non-sampling bias and nonresponse bias in section “[Non-response Bias](#).”

Whether non-sampling bias represents an important threat to a study depends on the research objective. If the researcher is interested in making generalizations to a particular target group, then non-sampling bias represents a severe threat. Thus, in the language of experimental research, if the research objective focuses on external validity, representative heterogeneous samples are required. However, if the

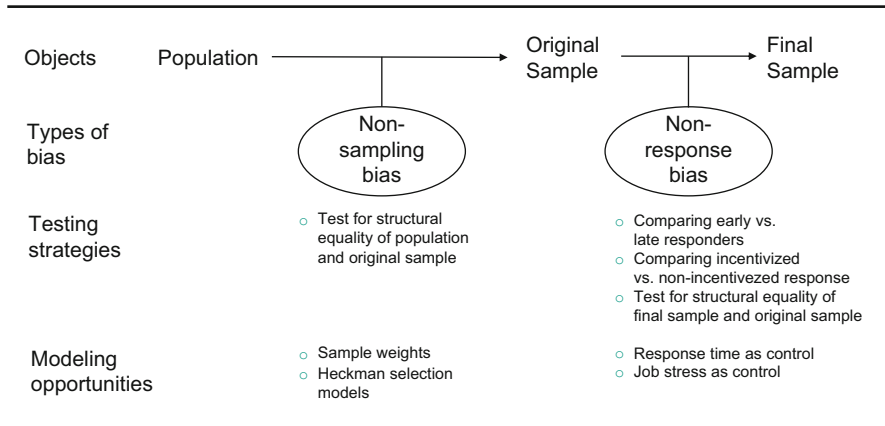


Fig. 3 Illustration of non-sampling and nonresponse bias. (Adapted from Homburg and Krohmer 2008)

researcher is interested in testing the veracity of proposed theoretical effects, non-sampling bias is a lesser issue. Stated differently, when the emphasis lies on internal validity, homogeneous samples can be adequate.

Short, Ketchen, and Palmer (2002) study the effects of non-sampling bias systematically. They link CEO duality, in which a firm's CEO concurrently chairs the board of directors, to a company's return on equity. The authors draw on different sampling frames, for which they observe substantially different effects: no effects in two cases, a positive effect in one case, and a negative effect in one case.

While sampling biases can have important consequences for research findings, the issue of sampling is often not systematically addressed. A review of studies published in leading management journals revealed that less than 40% of the studies included a discussion of representativeness (Short et al. 2002). Similarly, a review of marketing studies found that only 55% used an explicit sampling frame (Hulland et al. 2018). Finally, an important finding is that in general, samples chosen in social science are biased toward Western, educated, industrialized, rich, and democratized countries (Henrich et al. 2010).

Reasons for occurrence. Like resource constraints, the lack of suitable sampling frames can be a principal reason for non-sampling bias. For instance, while past consumer research could rely on telephone books for sampling frames, telephone books no longer validly resemble the population of interest (Iacobucci and Churchill 2010).

Similarly, for organizational studies, identification of adequate sampling frames can be challenging. For example, companies can be classified along numerous criteria (McKelvey 1975). In addition, the question arises regarding the organizational level at which the study should be conducted. A focus on strategic business units or product groups may present additional challenges for identifying adequate sampling frames (Sudman and Blair 1999).

Procedural remedies. The only procedural remedy is to base the sample on a sampling frame that is representative of the overall population. However, as mentioned, various reasons may obstruct knowledge of the overall population. Investigators may rely on sampling frames provided by data bases (e.g., COMPUSTAT), commercial mailing lists (e.g., Heide et al. 2014), or releases from Federal Statistical Offices regarding industry compositions (e.g., Vomberg et al. 2020). Nexis Uni (previously LexisNexis) may also serve to define the population for a study (Homburg et al. 2020). Nexis Uni has the advantage of including private companies, which are common for the business-to-business sector.

Statistical remedies. Statistical remedies can comprise methods to establish and to enhance representativeness. First, researchers can establish representativeness of their sample by comparing the structural equivalence of the original sample and the population (e.g., Vomberg et al. 2020). For instance, a χ^2 test may demonstrate the representativeness of the sample (Homburg et al. 2020).

Even if the distribution in the population is not known, researchers may compare the composition of two samples. For instance, Homburg et al. (2015b) compared the structural equivalence of samples obtained in two waves of data collection (years 1996 and 2013) in terms of industry sectors, sales volume, and number of employees.

Second, researchers may rely on Heckman selection models to control for non-sampling bias (Vomberg et al. 2020). Certo et al. (2016) provide a detailed discussion of Heckman selection models in the context of strategic management and also provide Stata code. Third, researchers may rely on sampling weights to enable the generalizability of their findings (Andreß et al. 2013; McElheran 2015; Raval 2020).

Non-response Bias

Conceptualization. In contrast to non-sampling bias, which refers to structural differences between the population and the original sample, non-response bias refers to structural differences between the original and the final sample (Fig. 3). Non-response bias may result from participants' failure to answer the survey (i.e., unit nonresponse) or from return of incomplete questionnaires (i.e., item nonresponse) (Klarmann 2008).

Analysis of reported response rates from leading management and organizational journals for the years 1975, 1985, and 1995 showed that the average response rate reported in these journals was 55.6% (with a large standard deviation of 19.7%) (Baruch 1999). However, responses have declined substantially over time – a trend that has been observed in various settings such as household studies (De Heer and De Leeuw 2002) or online surveys (Cook et al. 2000). Since the average US consumer receives more than 550 unsolicited surveys per year (compared with 50 to 100 for Germany, the UK, or France) (Iacobucci and Churchill 2010, p. 192), “oversampling” likely (partly) explains the dramatic reduction in response rates since Baruch's study. Today, response rates over 20% in organizational studies constitute the exception (Heide et al. 2007, 2014) and response rates around 10% represent the rule (Klarmann 2008).

In addition, response rates differ substantially between target groups. Top management studies in particular typically yield low response rates (Anseel et al. 2010).

However, participant drop-out does not necessarily threaten the results of a study. Non-response bias is a threat only when the reason for drop-out is related to the survey content (e.g., customers take part in a customer satisfaction survey depending on their satisfaction levels, or people with less discretionary time are less satisfied and less likely to respond to a satisfaction survey) and when sources that positively and negatively relate to survey content are not balanced (e.g., only unsatisfied customers do not reply in a customer satisfaction survey) (Thompson and Surface 2007).

Reasons for occurrence. Unit non-response is typically the result of respondents' refusal to participate in a survey. A review of factors that may drive unit non-response revealed personal factors, organizational factors, and survey-related factors (Klarmann 2008):

- *Personal factors:* Personal attitudes toward the survey (e.g., Helgeson et al. 2002; Rogelberg et al. 2001), involvement with the research topic (e.g., Groves et al. 2004), authorization (e.g., Tomaskovic-Devey et al. 1994; Gupta et al. 2000), and demographic criteria (e.g., Gannon et al. 1971; Gupta et al. 2000) have a strong impact on response rates.
- *Organizational factors:* Organizational factors such as industry profitability, dependence (i.e., subsidiary), and company size also influence the response rate (Tomaskovic-Devey et al. 1994).
- *Survey-related factors:* Number of contacts with participants (i.e., pre-notifications and reminder) (e.g., Yu and Cooper 1983), personalization of the survey (Yu and Cooper 1983), incentives (Church 1993; Yu and Cooper 1983; Yammarino et al. 1991), and length of the questionnaire (Yammarino et al. 1991) affect respondents' decisions to participate.

Procedural remedies. In the literature, four measures are discussed to diminish unit non-response bias (e.g., Klarmann 2008; Rogelberg and Stanton 2007; Anseel et al. 2010 for an in-depth review and evaluation of which techniques are effective for a particular target group):

- *Activities that increase the opportunity to participate in the survey:* These measures include (1) a deadline that can be determined by the participant (e.g., have the survey run for sufficient time so that vacation time does not impede response), (2) reminder notes, and (3) different modes of participation (e.g., written, online, per telephone)
- *Activities that emphasize the importance of the survey:* Emphasis can be achieved by (1) pre-notifying participants personally that they will receive a survey in the future, (2) reflecting participants' interests (e.g., an engaging title), (3) informing the participants about the survey goals.
- *Activities that decrease the perceived costs of the participation:* Researchers should (1) manage survey length and (2) carefully consider the survey design

- *Activities that raise the perceived utility of participation:* Researchers may provide incentives to participants. Common incentives in business-to-business contexts include social incentives (e.g., researchers donate of 10–15€ per participant for a social cause; Vomberg et al. 2020) and reports which outline the study results or benchmark a firm to sample averages; the latter two are more suitable incentives if participants are interested in the survey content. In business-to-consumer contexts payments (e.g., 5€ per participant; Homburg et al. 2019b), advance incentives (e.g., small gift included with the survey such as a pen), or a raffle (e.g., ten randomly chosen participants win a gift coupon from Amazon.com) represent commonly selected incentives. Thereby, the former two are likely effective even if participants are not involved with the survey content.

Statistical remedies. Researchers can use several different approaches to detect and to address a potential nonresponse bias (Rogelberg and Stanton 2007; Klarmann 2008). To establish that the final sample resembles the original sample, investigators can conduct a χ^2 goodness-of-fit test (Homburg et al. 2012b). If incentives are used for only a subset of respondents, then investigators could compare the answers of incentivized to non-incentivized participants. Researchers can also conduct follow-up interviews with non-respondents using a short version of the questionnaire (i.e., focal constructs) to determine the reasons for non-response (e.g., Groves 2006; Homburg et al. 2007). Finally, if researchers suspect that factors like job stress influence a participant's likelihood to respond, then they should also measure such variables and add them as control variables in their statistical models (Homburg et al. 2010).

Survey Research Process

In this section, we outline the general phases of survey research. Figure 4 shows that the typical survey research process starts with selection of research variables and ends with data analysis. Since we focus on the design of survey research in this section, we discuss questionnaire design in greater detail and comment only briefly on issues such as data analysis.

Selection of Research Variables

Which Relationships Are of Particular Interest?

In the first stage, the most important question is which relationships should be investigated. The answer is obviously determined by the research question the investigator wants to address. For instance, researchers might be interested in the role of formal and informal organizational elements for reacquiring customers (Vomberg et al. 2020). In this case, their conceptual model could contain formal reacquisition policies and an informal failure-tolerant organizational culture (Fig. 5).

Phase	Illustrative questions
1. Selection of research variables	Which relationships are of particular interest?
2. Selection of survey method	Should data be collected personally, by telephone, in written form or as an online survey? How can data be collected in an online context?
3. Questionnaire design	What questions should be asked? How should the questions be worded? In what order should the questions be asked?
4. Data collection	What is the ideal structure of the sample? How large should the sample be? How can we achieve a high response rate?
5. Measurement evaluation	To what extent do the survey questions measure What they are supposed to measure?
6. Data analysis	How are the examined phenomena related? How can the results be illustrated?

Fig. 4 Phases of the typical survey process

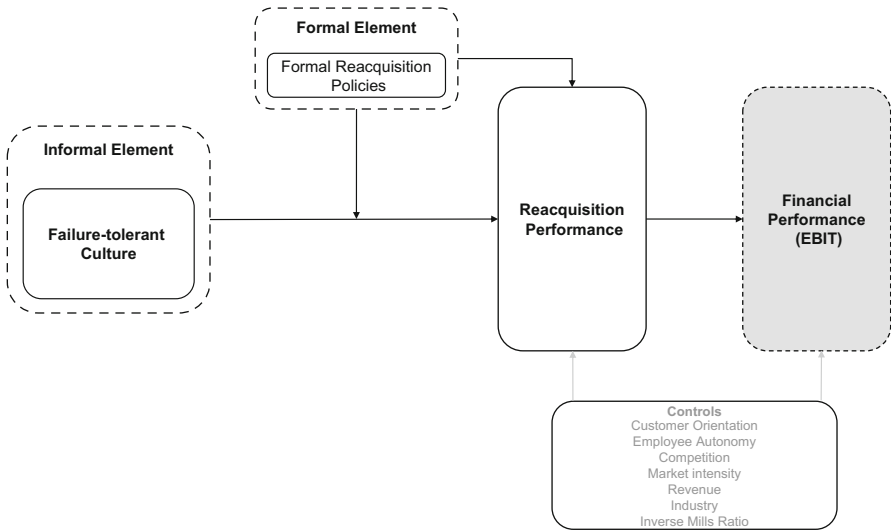


Fig. 5 Illustrative conceptual model (Vomberg et al. 2020, p. 121, reproduced with the kind permission of the publisher)

Importantly, the identification of variables requires familiarity with the research topic and is informed by prior research, conceptual considerations, or, for instance, in-depth interviews with practitioners. The conceptual model should

be as coherent and complete as possible, encompassing potential mediating and/or moderating effects if they are of interest for the subsequent analysis (e.g., Koschate-Fischer and Schulle (chapter ► [“Mediation Analysis in Experimental Research”](#)) in this handbook).

Selection of Survey Method

Should Data be Collected Personally, by Telephone, in Written Form, or as an Online Survey?

We discuss data collection modes only briefly because our focus is on written and online surveys. The personal interview takes place face-to-face between the interviewer and the interviewee. The interviewer reads out the items on the questionnaire to the interviewees and collects their answers. The telephone interview can be a short, cold-call interview or a planned interview. The written survey usually involves sending questionnaires to the respondent by mail, with the respondent then returning the completed questionnaire. The online survey can be conducted in various ways – as a questionnaire attached to an e-mail or as a link provided on a website, in an e-mail, or in a QR-code.

Table 1 which is based on Homburg (2020) systematically compares the appropriateness of the four survey methods with regard to the (1) suitability for the object of study, (2) extent of data, (3) quality of data, and (4) length and costs of the project. These general evaluations may differ according to the survey context. In addition, the four forms of survey method likely differ in the sources of biases that may relate with them. Usually, personal interviews and telephone surveys are more prone to social desirability bias or acquiescence compared to self-administered questionnaires (i.e., written and online surveys) (Krosnick 1999; MacKenzie and Podsakoff 2012). In telephone interviews, midpoint response styles are less common than in mail or online surveys. In addition, telephone surveys tend toward acquiescence. Compared to mail surveys, online surveys display fewer disacquiescence and extreme response styles (Weijters et al. 2008; section [“Response Styles”](#)).

How Can Data Be Collected in an Online Context?

Current trends in consumer research suggest that researchers increasingly rely on crowdsourcing platforms to obtain convenient and inexpensive samples. Mechanical Turk (MTurk), which was launched by Amazon in 2005, currently is the prevalent option. On MTurk, researchers act as employers and hire and compensate workers for participation in surveys, giving MTurk several likely advantages. Since incentives are comparably low on MTurk, researchers can conduct studies at lower costs without necessarily compromising data quality. In addition, participants likely have strong incentives to fill in the survey with due diligence, since they receive compensation only after passing screening and attention questions. Finally, MTurk allows researchers to implement their studies quickly (e.g., Chandler and Paolacci 2017; Levay et al. 2016).

However, skepticism regarding the use of MTurk is increasing. First, while MTurk samples are likely more heterogeneous than student samples, they are not

Table 1 Comparison of different survey methods

Criteria	Standardized verbal survey	Standardized telephone survey	Standardized written survey	Standardized online survey
Suitability for the object of study				
Possibility to explain a complex issue	Very good	Good	Rather low	Rather low
Possibility to use complex scales and filter questions in the questionnaire	Only when computer-aided	Only when computer-aided	Low	Very good
Possibility to demonstrate trial product samples	Very good	Rather low	Rather low	Rather low
Possibility to include multimedia (e.g., video and sound)	Rather good	Medium	Low	Very good
Extent of data				
Subjective length of the survey evaluated by the interviewee	Short	Medium	Long	Very long
Possibility to question a large sample	Low	Medium	Very high	High
Response rate	High	Medium	Low	Low
Risk of aborting the interview	Low	Medium	High	Very high
Quality of data				
Possibility of inquiries when comprehension problems occur	Very good	Good	Very low	Very low
Possibility of distortion due to the social interaction with the interviewer	Very high	High	None	None
Possibility to modify the research instruments during the field phase	Very good	Good	Low	Good
Length and costs of the project				
Length of the field phase	Medium	Short	Long	Long
Costs	Very high	High	Low	Low

necessarily representative of the overall consumer population. Second, participants on MTurk can be experienced survey takers who answer differently than less experienced survey respondents. Third, participants on MTurk can misrepresent themselves to participate in attractive studies with high payouts. However, misrepresentation is a general threat to online research (Goodman et al. 2013; Goodman and Paolacci 2017; Hulland and Miller 2018; Wessling et al. 2017).

TurkPrime and Prolific Academic (ProA) have recently emerged as alternatives to MTurk. TurkPrime is a website that utilizes MTurk but claims to overcome potential disadvantages of MTurk (e.g., participant misrepresentation) (Litman et al. 2017).

Graduate students from Oxford and Sheffield Universities launched ProA (<http://www.prolific.co>) in 2014. ProA provides several demographic details about its participant pool. Peer et al. (2017) who compare ProA with MTurk note that ProA participants are less experienced with research paradigms and provide more honest answers than MTurk participants. Marketing researchers increasingly rely on ProA for their studies (e.g., Castelo et al. 2019; Hagen 2020).

Qualtrics, Survey Sampling, Inc. (SSI), or Critical Mix, whose samples are likely more representative than MTurk samples, are other potential ways to collect data. However, these platforms are also more expensive. Finally, for short surveys (up to 10 questions) Google Surveys may offer a valid option. On Google Surveys, respondents take part because they want to read an article, lowering the likelihood of misrepresentation threats (Wessling et al. 2017; Hulland and Miller 2018).

Questionnaire Design

The design of the questionnaire usually represents the crucial stage in the collection of survey data. Mistakes that occur in this stage are normally hard to revise in later stages. Figure 6 outlines the phases of questionnaire design and presents examples of questions during the different phases.

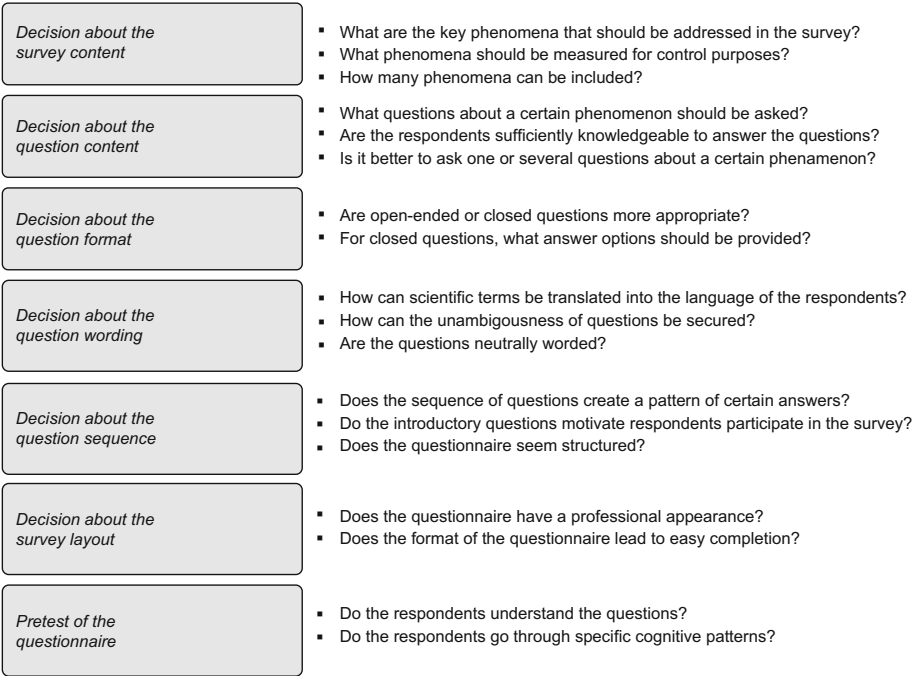


Fig. 6 Phases of the typical questionnaire design process

Decision About the Survey Content

Three fundamental questions determine survey content: What key phenomena should the survey address? What phenomena should be measured for control purposes? How many phenomena can be included in the survey?

What Are the Key Phenomena that Should Be Addressed in the Survey?

The key phenomena are provided by the research questions and the formulated hypotheses. Drawing on the conceptual framework (section “[Selection of Research Variables](#)”), investigators decide on the phenomena/constructs that should be included in the survey. In our example (Fig. 5), investigators would include variables that capture formal reacquisition policies, failure tolerance, and reacquisition performance. If financial performance cannot be obtained from archival data bases, then measures of financial performance should also be collected.

What Phenomena Should Be Measured for Control Purposes?

Besides including variables for substantive reasons, researchers should include theoretically motivated control variables to reduce “noise” in the independent and dependent variables. More importantly, however, survey researchers rely on control variables to account for non-randomness in their data. In contrast to experimental data, survey data are a form of observational data, and thus effects observed from survey data might be correlational and not causal in nature. To increase survey researchers’ ability to infer causality, control variables serve to rule out rival explanations (section “[Endogeneity in Survey Research](#)”). For instance, in our example (Fig. 5), customer orientation may represent an important control variable. Theory suggests that customer-oriented companies might be more successful in winning customers back (i.e., customer orientation may affect the dependent variable) and also are more likely to have formal reacquisition policies in place (i.e., customer orientation may influence the independent variable). However, additional variables have been included to account for further potential differences.

To systematically collect control variables, researchers should review prior literature, follow theoretical considerations, and rely on their own plausibility considerations. They should then store the identified variables in a long list, such as an Excel sheet containing potentially relevant variables. However, in many cases, researchers end up with a list that captures too many variables to be assessed in one questionnaire and have to make a trade-off between questionnaire length and the completeness of the variables.

How Many Phenomena Can Be Included?

Consideration of questionnaire length is essential to reducing potential biases. Excessive length of the questionnaire can lead to respondent fatigue, resulting in careless and biased responses (MacKenzie and Podsakoff 2012) and contributing to an acquiescence response style (section “[Response Styles](#)”). In addition, long questionnaires can lead to survey non-response (section “[Non-response Bias](#)”), which may threaten the representativeness of the survey or lead to sample sizes that are too small for statistical analysis. As rules of thumb, we recommended the following:

- Telephone interview: 20–30 mins (planned interview; less for a cold call)
- Pop-up online survey: 25 questions
- Written survey: 100 questions

Researchers should also carefully evaluate options discussed in section “[Sources of Systematic Errors in Survey Research: Measurement Errors](#)” on how to control for biasing effects, such as using dedicated variables to account for potential CMB through the measured latent factor approach. However, this approach requires that the questionnaire becomes longer and that other substantially important variables be dropped from the questionnaire. Therefore, when designing their surveys, researchers need to critically reflect which biases are likely to present threats to the validity of their findings.

However, we point out that these rules of thumb are general, as situational factors also influence the length of questionnaires. For instance, questionnaires for high-level corporate decision makers need to be shorter, whereas surveys of lower-level employees might be longer. In addition, whether participants are willing to answer longer questionnaires depends heavily on their involvement with the research topic. If participants consider the survey interesting, they may be willing to answer longer questionnaires (section “[Nonresponse Bias](#)” discusses measures to stimulate involvement).

Decisions About the Question Content

A next critical step is to decide on the question content. Typical questions in this step are: What questions about a certain phenomenon should be asked? Are the respondents sufficiently knowledgeable to answer the questions? Is it better to ask one question or more about a certain phenomenon? Which questions should be selected?

What Questions about a Certain Phenomenon Should Be Asked?

To determine which questions are appropriate for certain phenomena, researchers first need to conceptually define their constructs of interest. These definitions logically precede the operationalization of the construct – that is, the process of coming up with items for their measurement.

The next important decision is whether to rely on new or established scales to measure the constructs. For many constructs, various established scales already exist in the literature. Researchers can find scales in the following ways:

- Manuals such as the *Handbook of Marketing Scales* (Bearden and Netemeyer 1999) provide an overview of established scales.
- The website inn.theorizeit.org allows online search for established measurement scales
- Articles published in leading journals typically provide the items for the scales they used (e.g., Homburg et al. 2011).

Researchers can systematically collect these established questions with appropriate references in an Excel sheet.

An important advantage of relying on existing scales is that they enjoy wide acceptance in the academic community because prior research has rigorously

demonstrated their psychometric properties (e.g., Homburg et al. 2015a). In addition, development of original questions can sometimes lead to problems in evaluating their reliability and validity in later stages (section “[Measurement Evaluation](#)”), as even after rigorous pretesting, newly developed scales may not display required psychometric properties in this handbook).

However, a middle way is possible: researchers can combine newly developed items with established items to better fit their research context. In addition, researchers can adapt established items to the context of their study. For instance, Kumar et al. (1995) relied on established items from Price and Mueller (1986) but adapted them to context of their study (e.g., established item: “To what extent are you fairly rewarded for the amount of effort you put forth” vs. adapted item: “How fair are your firm’s outcomes and earnings compared to the contributions we make to this supplier’s marketing effort”).

Are the Respondents Sufficiently Knowledgeable to Answer the Questions?

When selecting questions for the constructs of interest, researchers need to have their target respondent characteristics in mind (e.g., educational background). Researchers should especially evaluate whether respondents are sufficiently knowledgeable to respond to the questions – a pivotal concern in key informant studies (section “[Key Informant Bias](#)”). If respondents lack the ability to answer the particular questions, they may take on an acquiescence response style (section “[Response Styles](#)”).

Is it Better to Ask One Question or Several about a Certain Phenomenon?

When deciding about the questions’ content, researchers need to determine the number of questions they intend to ask per variable because adding redundant or unnecessary questions increases the questionnaire’s length, perhaps squeezing out questions to collect information on other constructs or evoking the respondent’s refusal to participate in the survey (Bergkvist and Rossiter 2007). However, relying on too few questions may result in problems regarding the reliability or validity of the measurement (section “[Reliability and Validity: Fundamentals](#)”).

In the marketing literature, debate is ongoing as to whether and when single items can be used (Bergkvist and Rossiter 2007; Diamantopoulos et al. 2012). Usually, single items are sufficient when a construct can be precisely captured (e.g., demographic information, revenues). However, in the case of less concrete constructs (e.g., customer satisfaction, company culture, and leadership styles) multiple-item scales offer several benefits. First, multiple-item scales help to control random measurement error. Addition of further items to the measurement of one construct typically increases the overall reliability of the scale (section “[Measurement Theory](#)”). Second, multiple-item measures are more likely to capture all facets of the construct of interest (Baumgartner and Homburg 1996). For instance, the construct transformational leadership, which captures leader behaviors that make followers more aware of the importance of their job and that inspire employees to strive for the benefit of the company, is composed of facets such as “identifying and articulating a vision,” “providing an appropriate model,” or intellectually stimulating employees, thus requiring multiple items for its measurement (Podsakoff et al. 1990). Third,

multiple-item measures enable evaluation of the psychometrics properties for the respective scales (section “[Measurement Evaluation](#),” in this handbook).

In the following, we provide analytical evidence that increasing the number of items can enhance the overall reliability of a construct (Moosbrugger 2008). Equation 3 formally defines the reliability of a construct (x) measured with a single item (e.g., the researcher measures customer satisfaction with a single question).

$$\text{Rel}(x) = \frac{\text{Var}(t)}{\text{Var}(x)} = \frac{\text{Var}(t)}{\text{Var}(t + e)}.$$

(3)

The observed variance of the construct x (customer satisfaction) is split into a true value t and a random error component e . Recall that reliability refers to random but not to systematic errors (section “[Reliability and Validity: Fundamentals](#)”). Therefore, we do not include a systematic error in Eq. 3. The resulting reliability measure from Eq. 3 ranges from 0 to 1, with larger values indicating higher levels of reliability.

When measuring a construct with multiple items ($x_1, x_2, \dots, x_k$), the overall reliability of the scale (e.g., reliability of the overall customer satisfaction measure) can be calculated according to Eq. 4. For simplicity, Eq. 4 assumes that each individual item has the same individual reliability $\text{Rel}(x)$.

$$\text{Rel}_{k \times x} = \frac{k \times \text{Rel}(x)}{1 + (k - 1) \times \text{Rel}(x)}.$$

(4)

Table 2 is based on Eq. 4 and demonstrates that the overall reliability of a construct increases with the number of items. To be more precise, Table 2 demonstrates the reliability of a scale depending on the number of items and individual reliability of an item (i.e., $\text{Rel}(x) = 0.40$ and $\text{Rel}(x) = 0.60$). For instance, if an individual item has a reliability of $\text{Rel} = 0.40$ and the researcher uses this single item

Table 2 Illustration of how number of items increases construct reliability

# Items	Scale reliability	
	Individual item reliability	
	0.40	0.60
1	0.40	0.60
2	0.57	0.75
3	0.67	0.82
4	0.73	0.86
5	0.77	0.88
6	0.80	0.90
7	0.82	0.91
8	0.84	0.92
9	0.86	0.93
10	0.87	0.94

to measure a construct, then the overall reliability of the scale is also $Rel_{1 \times x} = 0.40$. However, if the investigator asks five items to measure one construct (i.e., the questionnaire contains five different questions that all measure customer satisfaction) and each item has an individual reliability of $Rel(x) = 0.40$, the overall reliability of the scale increases to $Rel_{5 \times x} = 0.77$ – a result that clearly demonstrates the advantage of using multiple items. In addition, Table 2 demonstrates that scale reliability does not increase linearly with the number of items (i.e., moving from one to two items increases scale reliability more than moving from nine to ten items).

Diamantopoulos et al. (2012) replicate the analytical evidence in a simulation study and demonstrate that in most instances multiple-item scales outperform single-item scales. Situations in which single items outperform multiple items are not realistic in applied research, as researchers would not have a priori knowledge about which item is most adequate in the specific context.

Against these observations, we recommend relying on single items for concrete constructs (e.g., Carson and Ghosh 2019) but using multiple items for complex constructs (Diamantopoulos et al. 2012; Hulland et al. 2018). In the case of multiple items, as a general rule, researchers should select between four to six items (Table 2). In addition, to avoid reactance from respondents when evaluating similar questions, in the beginning of the survey researchers should point out the necessity of repeating similar questions.

Which Questions Should Be Selected?

A final and important decision is which questions should be selected. The domain-sampling model addresses this question theoretically. The domain-sampling model postulates that a given construct has a broad universe of possible items, behaviors, and responses that can serve as its observable markers or indicators (Nunnally 1967). Relying on concepts from sample selection theory (e.g., the concept of representativeness), the domain sampling model proposes that a valid representation of a construct is achieved when representative items are selected. Representative items can be identified through random sampling – that is, researchers randomly pick items from their list of items that could be used to measure the construct. However, when a given domain is adequately specified (e.g., through prior research or conceptual considerations), the researcher can deliberately select (rather than sample) a set of items (Little et al. 1999).

Decision About the Question Format

The important issue in this step is whether open-ended or closed questions are more appropriate, and for closed questions, what answer options should be provided?

Are Open-Ended or Closed Questions More Appropriate?

Open-ended questions allow participants to freely articulate their thoughts and opinions, while closed questions require participants to choose answers from a given list. Closed questions can have important advantages particularly for written and online surveys. First, the meaning of provided answers to closed questions might be clearer to the researcher than answers to open-ended questions, which likely require subjective interpretation. Relatedly, many statistical techniques

(e.g., regression analysis) require quantitative and structured data (however, the automated analysis of qualitative data is evolving; e.g., Humphreys (chapter ► [“Automated Text Analysis”](#)) in this handbook). Second, open-ended questions require great effort on the part of the respondent and may lead to participant refusal. Third and as a consequence of the aforementioned, researchers typically are able to ask more closed questions than open-ended questions – an aspect particularly relevant in organizational studies, for which participants are hard to recruit.

However, open-ended questions offer two potential benefits. First, participants can provide unusual and spontaneous answers to open-ended questions whereas their ability to respond is constrained by closed questions – if the investigator’s list of questions and answer options omits important aspects, participants cannot deliver insights in this regard. Second, open-ended questions require less in-depth knowledge than closed questions, and participants can provide insights into topics the investigator has not considered.

However, we question whether these potential benefits arise for written or online surveys. First, we doubt that in all domains participants can better evaluate than the investigator what could be relevant for the research question (Schuman and Presser 1981). Second, investigators can rely on pretesting to discover and fix omissions in their closed questions. Schuman and Presser (1979) demonstrate that answers to closed and open-ended questions are comparable when closed questions are adapted to insights from pretests (section [“Pre-test of the Questionnaire”](#)).

Our experience mirrors this discussion. We have noted that the quality of answers to open-ended questions in written surveys tends to be low. Consequently, we recommend focusing on closed questions in written and online surveys. However, we recommend that if possible, researchers change scales between dependent and independent variables to alleviate potential CMB (section [“Common Method Bias”](#)). We also recommend including an open-ended question at the end of the questionnaire allowing participants to provide further insights (e.g., “Are there additional issues you want to address?”).

Relatedly, including an open-ended question has an additional advantage: it lowers the probability that respondents engage in response substitution, which occurs when respondents want to express attitudes and beliefs that the researcher has not asked about. Informing respondents at the beginning of the questionnaire that they will have an opportunity to express any other thoughts in an open-ended format can reduce the biasing effects of response substitution (Gal and Rucker 2011).

For Closed Questions, What Answer Options Should Be Provided?

When selecting answer categories for closed questions, researchers have various options. For instance, closed questions can be multichotomous (e.g., automotive industry, financial industry, consumer durables), dichotomous (e.g., purchase vs. no purchase), or measured on a scale (e.g., Likert scale, frequency scale). When a Likert scale is chosen, a frequently asked question is how many response categories should be provided. The empirical evidence on this matter demonstrates that reliability tends to increase with an increasing number of scale options. However,

the marginal utility also tends to decrease. For instance, the quality of the provided answers is better for six than for four response categories (Preston and Colman 2000), and indicators with more response categories tend to display higher reliability (Alwin and Krosnick 1991). However, for more than seven response categories very few additional gains are observed. Thus, we recommend relying on five to seven answer categories.

In addition, participants can adapt their responses to seemingly arbitrary choices of scale labeling. For instance, Schwarz et al. (1991a) relied on two different labels for the same scale. In the first case, the scale ranged from 0 (“not at all successful”) to 10 (“extremely successful”). In the second, the anchor values ranged from -5 (“not at all successful”) to $+5$ (“extremely successful”). The result was that 34% of the respondents selected values between -5 and 0 on the scale from -5 to $+5$, but only 13% chose the equivalent values between 0 and 5 on a scale from 0 to 10. Since such behaviors are hard to anticipate, this example emphasizes the need for conducting pretests (section “[Pretest of the Questionnaire](#)”).

Decisions about the Question Wording

Ambiguous question formulations can obviously bias the responses to questionnaires. If respondents do not comprehend the question, they cannot provide adequate information (Baumgartner et al. 2018). Thus, to ensure comprehension, question wording should be (1) simple, (2) unambiguous, and (3) neutral.

Simplicity is important, as complex items likely encourage respondents to develop their own idiosyncratic understanding of questions and/or to use biased response styles. In this regard, researchers need to decide whether to rely on item reversals. Item reversal can be achieved in two ways (Baumgartner et al. 2018; Weijters and Baumgartner 2012; Weijters et al. 2009):

- *Negations*: using “not” (“I see myself as someone who is. . .” talkative vs. *not* talkative), affixal negation of adjectives (e.g., *dishonest*), or using negative adjectives or adverbs (e.g., *seldom*) or the negation of a noun with “no.”
- *Polar opposite items*: e.g., “I enjoy taking chances in buying new products” versus “I am very cautious in buying new products” or “talkative” versus “quiet.”

We recommend that researchers avoid negations and use polar opposite items carefully. Reversed items can offer advantages: they can inhibit stylistic responding (e.g., acquiescence bias; section “[Response Styles](#)”), act as cognitive “speed bumps,” and disrupt non-substantive response behaviors. However, and more pressingly, they might also lead to respondent confusion, lowering the quality of the collected data (e.g., Baumgartner et al. 2018; Weijters et al. 2009; Wong et al. 2003).

Simplicity can also be achieved when investigators

- Use rather short sentences and refrain from pyramiding (e.g., through relative clauses), as pyramiding or longer sentences can lead to complexity
- Avoid unnecessary variations in the format and structure of the questions
- Translate scientific language into the participants’ language

- Avoid requiring participants to do computations: for example, instead of directly asking “how much do you spend on average in a focal supermarket per month,” investigators should ask two simpler questions (e.g., “how often do you go to the focal supermarket per month?” and “How much do you spend on average per shopping trip at the focal supermarket?”)

Unambiguousness also avoids biases. Item ambiguity impairs the respondent’s ability to comprehend the question’s meaning and clearly undermines the respondent’s ability to provide accurate answers. When respondents are uncertain about how to respond to the item’s content, systematic response patterns are likely evoked and respondents are more likely to rely on their own implicit theories.

Strategies to avoid item ambiguity include the use of precise and concise language. For instance, universal expressions (e.g., “all,” “always”) or vague quantifiers (e.g., “often,” “many”) mean different things to different respondents. Thus, expressions such as “I read many books” would be less valid than “I read ten books a year” (Johnson 2004). Churchill and Iacobucci (2005, p. 248) summarize further aspects which need to be considered in this regard (e.g., “about” could mean “somewhere near” or “almost,” “like” distracts the attention of participants to the specific examples, and answers to the word “often” largely depend on the respondent’s frame of reference).

Importantly, researchers should use the same words for the same issues. In addition, they should clearly define terms that respondents might interpret differently. For instance, in research on customer reacquisition management in business-to-business markets, in-depth interviews revealed that customer defections can be defined differently. For some companies, only customers who completely stopped purchasing qualified as defected. For other companies, customers that had lowered their purchasing volume represented defectors. Thus, in designing a subsequent questionnaire, the researchers defined customer defection in the beginning of the questionnaire to avoid ambiguity (Homburg et al. 2019a).

Neutrality is also important when formulating questions. To achieve neutrality, researchers should avoid suggestive formulations and anticipations of answers (e.g., “Do you also agree?”). Finally, researchers should avoid loaded words (e.g., market economy vs. capitalism) that are likely to bias participants’ answers.

Decisions about the Question Sequence

Does the Sequence of Questions Foster a Pattern of Certain Answers?

As noted earlier, participants may form their opinions while answering the questionnaire (section “[Psychology of Survey Response](#)”). That is, answers provided to initial questions may activate information to answer questions at later points in the questionnaire (Schwarz 2003). Thus, researchers need to avoid generating survey data that yields only “self-generated validity” (Feldman and Lynch 1988, p. 421).

A prominent example from the field of political science that has been repeatedly affirmed since 1948 illustrates the effects of question sequence. In this example, participants were asked two questions:

- *Question A*: Do you think the United States should let Communist newspaper reporters from other countries come in here and send back to their papers the news as they see it?
- *Question B*: Do you think a Communist country like Russia should let American newspaper reporters come in and send back to America the news as they see it?

In randomized experiments, the share of “yes” answers pertaining to *Question A* largely depended on the order of the two questions. The group who saw *Question A* and then *Question B* agreed to question *Question A* in 44% of the cases. However, the group that was asked in reversed sequence (first *Question B* then *Question A*) had higher agreement with *Question A* (70% agreement) (Schuman et al. 1983; Schwarz et al. 1991b report similar examples).

These sequence effects also become important in customer satisfaction surveys. A central decision in customer satisfaction surveys usually is at which point the investigator should ask participants to rate their overall level of satisfaction (e.g., “Overall, please indicate how satisfied you are with the company”). Should the rating be solicited before or after asking specific customer satisfaction questions (e.g., “Please indicate your satisfaction regarding the products,” “Please indicate your satisfaction regarding the after-sales service”)?

Researchers observed that depending on the sequence, different psychological processes set in. If overall customer satisfaction is measured at the beginning of the survey, emotions typically dominate the assessment (e.g., gut feeling). However, if total customer satisfaction is measured at the end of the survey, the same question will trigger cognitive rather than emotional processes. Particularly in this case, the answers to the overall customer satisfaction question will likely depend on the replies to the specific customer satisfaction responses participants provided before.

Which order is the most appropriate largely depends on the research context. Researchers may measure overall customer satisfaction at the beginning (end) of the questionnaire for low (high) involvement products. As emotions (cognitions) are likely to be an important factor when purchasing low (high) involvement products, a more emotional (cognitive) response to overall customer satisfaction might be more insightful.

The literature discusses some measures researchers can employ to reduce the effects of question sequence. One approach could be to randomize the question order, which can easily be achieved in online surveys: researchers present the same questions in different orders to participants. Although this approach might seem intuitively appealing, the downside is that it cannot prevent order effects from occurring on the individual level. In addition, randomization may disrupt the logical order of the questions. An alternative might be to simply tell respondents that the order of presentation is random and therefore of no relevance (Krosnick et al. 1990).

The inclusion of buffer questions might also help to reduce sequence effects. For instance, researchers can include buffer questions between items they expect to evoke a large sequence effect – a tactic that likely lowers artificial correlations among items (Weijters et al. 2009).

Do the Introductory Questions Motivate Respondents to Participate in the Survey? Does the Questionnaire Seem Structured?

In general, we recommend that researchers start with an interesting opening question. This question should motivate participants to take part in the survey, and its importance increases in light of the declining response rates for questionnaires (section “[Non-sampling Bias](#)”). After the opening question, researchers should move from general to specific questions and ask sensitive questions later in the survey. Rapport is established as respondents answer general and nonthreatening questions early in the survey. Finally, when researchers cannot access different sources for evaluating independent and dependent variables, they should first assess the dependent and then the independent variables. Thereby, they lower potential demand effects (section “[Common Method Bias](#)”).

Decisions About the Survey Layout and Pretest of the Questionnaire

Does the Questionnaire Have a Professional Appearance? Does the Format of the Questionnaire Lead to Easy Completion?

The layout of the questionnaire is a critical success factor. Participants perceive the questionnaire to represent the net of the asked questions, layout and design, and logical structure and architecture. All these aspects may affect the effort participants put into answering the survey and thus affect the likelihood of biases to occur (Presser et al. 2004).

In general, the questionnaire “should look as sharp as your résumé” (Iacobucci and Churchill 2010, p. 221). A clean appearance signals interest for the topic and also emphasizes the researcher’s trustworthiness. A questionnaire’s layout may also create the impression of short processing time and thereby increase response rates, lowering the threat of non-response bias (section “[Non-response Bias](#)”). Finally, the instructions should be clearly articulated and the question flow should be evident and supportive.

Pretest of the Questionnaire

Do Respondents Understand the Questions? Do Respondents Engage in Specific Cognitive Patterns?

While textbooks on the development of a survey are plentiful, there are “no silver bullets of questionnaire design” (Schwarz 2003, p. 593), and even experts struggle to anticipate all potential difficulties that arise when participants address the survey questions. Thus, “there is no substitute for in-depth pretesting” (Weijters and Baumgartner 2012, p. 565) and “*all* researchers should use pretests prior to running their main studies” (Hulland et al. 2018, p. 104, emphasis in original).

The goals an investigator may want to achieve when running a pretest are varied and include the following:

- Identification of ambiguous questions and problems of understanding
- Detection of incompleteness of answer categories
- Identification of potential information gaps of respondents
- Verification of the time required to complete the questionnaire

- A preliminary indication of validity and reliability of the measurement (but taking the pretest sample size into account)

Three methods usually pertain to pretests. First, the debriefing method permits pre-testers to complete the questionnaire and then discuss potential issues (for instance, online surveys allow participants to make notes that can be discussed afterwards). Online pretests allow investigators to track pre-tester behaviors, which they can discuss in debriefing meetings. For instance, they could track response time or employ eye-tracking software to identify problematic survey sections (Baumgartner et al. 2018). Second, the protocol method allows pre-testers to raise questions when filling out the questionnaire. Third, the think-aloud method (also referred to as cognitive interviews) require pre-testers to verbalize their thoughts when filling out the questionnaire (Ericsson and Simon 1980; Presser et al. 2004; Weijters et al. 2009).

Data Collection

As our focus is on the design aspect of survey research, we do not discuss in detail various data collection methods but confine our discussion to the importance of sample structure, sample size, and sample response rate. In this handbook, Bornemann and Hattula (chapter ► “Experiments in Market Research”) provide a more detailed discussion of sampling procedures.

What Is the Ideal Structure of the Sample?

Sampling designs can be divided into probability and non-probability samples. Probability samples are random samples. For instance, a simple random sample assumes that observations are selected by chance from the population. While most statistical tests assume probability sampling, applied researchers often rely on non-probability sampling since random samples are often hard to obtain (Short et al. 2002). As already mentioned, in many situations the population of interest is hard to define (section “Non-sampling Bias”), which limits investigators’ ability to draw random samples.

Literature distinguishes four principal ways of non-probability sampling. First, researchers might rely on convenience samples – that is, they select a sample on the basis of accessibility. Convenience samples are the most typical sampling form in marketing and management research (Short, Ketchen, and Palmer (Short et al. 2002) report 42%, and Hulland et al. (2018) report 43%). However, convenience samples are likely to be unrepresentative of the overall population. Importantly, in consumer research debate is ongoing about the use of a particular type of convenience sample – student samples (Peterson 2001).

Second, researchers may apply quota sampling where units are drawn to approximate known proportions found in a population. Third, investigators may rely on snowball sampling, in which researchers identify a few participants from the

population of interest and ask them to forward the questionnaire. This technique is useful when determining the population of interest is highly challenging (e.g., as when market research focuses on extreme sports). Fourth, sampling may be based on typicality. Here, researchers do not focus on representative participants but on participants from whom they expect valuable input. This sampling approach is often used during pretests (section “[Pretest of the Questionnaire](#)”; Iacobucci and Churchill 2010; Short et al. 2002).

How Large Should the Sample Be?

The question of sample size is hard to answer without considering the specific research setting. A rule of thumb for regression analysis is to rely on 10 observations per parameter (Hair et al. 2010). However, in applied research projects, this ratio is usually larger. For instance, the ratio of sample size to variables could be 70.63 (Short et al. 2002).

Small samples can have important disadvantages. Smaller samples can lead to less reliable test statistics, as standard errors may increase. In addition, small samples may limit the types of statistical techniques that can be applied effectively. However, large samples can also be problematic, as negligible effects might become statistically significant owing to high statistical power.

Finally and importantly, no clear relationship exists between sample size and the sample’s representativeness. Opinion polls have revealed that if researchers invest substantial effort in acquiring large samples, representativeness of the final sample might suffer. Therefore, sampling should prioritize representativeness over sample size (Assael and Keon 1982; Krosnick 1999; Rogelberg and Stanton 2007).

How Can we Achieve a High Response Rate?

In section “[Non-response Bias](#),” we discussed several techniques for countering potential non-response bias. As the same techniques should help to increase the response rate (Anseel et al. 2010), we do not describe these actions here. However, we note that the size of a response rate does not indicate the representativeness of a sample. In some instances, lower response rates can lead to more accurate results than higher response rates (Visser et al. 1996). Recent evidence confirms this observation and emphasizes that the assumed positive relationship between response rate and survey quality does not always hold (The American Association for Public Opinion Research 2016).

Measurement Evaluation

To What Extent Do the Survey Questions Measure What they Are Supposed to Measure?

After the questionnaires have been returned to the researcher, the quality of their measurement can be evaluated. In-depth discussions of how survey researchers can demonstrate the reliability and validity of their constructs appear elsewhere in this handbook (chapter ► “[Structural Equation Modeling](#)” by Baumgartner and Weijters).

In the following, we briefly mention which statistics researchers typically report to provide initial guidance. All of the presented techniques require researchers' use of multiple-item scales (section [“Decisions About the Question Content”](#)).

Researchers can demonstrate the reliability of their constructs by reporting Cronbach's alpha and/or composite reliability values (both have threshold values of >0.70). Both estimate the squared correlation between a construct and an unweighted sum of its items; composite reliability represents a generalization of Cronbach's alpha because it does not assume equal loading across items.

Through confirmatory factor analysis, investigators can demonstrate convergent and discriminant validity. Convergent validity (which affirms that the items correlate with other concepts they should be related to) can be established for individual items by squaring the individual standardized factor loadings (threshold value: >0.40). At the construct level, researchers calculate the average variance extracted (threshold value: >0.50) (Baumgartner and Weijters (chapter [▶ “Structural Equation Modeling”](#)) offer more flexible cut-off values).

Researchers can rely on the Fornell-Larcker criterion (1981) to establish discriminant validity (which affirms the items are uncorrelated with concepts they are not related to). For each pair of constructs, the square root of the average variance extracted for each construct needs to exceed the correlation between them.

Data Analysis

How Are the Examined Phenomena Related? How Can the Results Be Illustrated?

Analysis of the acquired data is the final step in the typical survey research process. We mention this step only briefly for two reasons. First, important decisions in survey research precede the statistical analyses. Hence, the careful design of the questionnaire is a necessary condition for the empirical analysis to yield valid results. Second, many statistical techniques can be used to analyze survey data and their in-depth discussions appear in designated method chapters of this handbook: most commonly, researchers rely on regression analysis (chapter [▶ “Regression Analysis”](#) by Skiera et al.), structural equation modeling (Baumgartner and Weijters 2017), partial least squares structural equation modeling (chapter [▶ “Partial Least Squares Structural Equation Modeling”](#) by Sarstedt et al.), or analysis of variance (chapter [▶ “Analysis of Variance”](#) Landwehr).

Endogeneity in Survey Research

Addressing endogeneity concerns is typically important when investigators rely on observational data such as survey data (Sande and Ghosh 2018). For instance, the study of Huang and Sudhir (2021) demonstrates the need for survey researchers to address endogeneity concerns. Their study shows that not accounting for endogeneity leads to an underestimation of the true effect of service satisfaction on

customer loyalty. Such underestimation likely leads to less than optimal investment decisions in business practice. While an in-depth discussion of endogeneity is provided elsewhere in this handbook (chapter ► [“Dealing with Endogeneity: A Nontechnical Guide for Marketing Researchers”](#) by Ebbes et al.), we highlight several aspects that might be particularly important for survey researchers.

Conceptually, endogeneity concerns refer to alternative or rival causal explanations of the findings. Statistically, endogeneity concerns arise if the explanatory variables correlate with the error terms, for example in regression analyses. Literature distinguishes three sources of endogeneity: (1) omitted variables, (2) measurement error, and (3) simultaneity (e.g., Cameron and Trivedi 2005; Kennedy 2008; Rossi 2014).

First, with regard to omitted variables, survey research has an important advantage over secondary data. In principle, survey researchers can rule out omitted variable bias by including control variables (section [“Decision About the Survey Content”](#)). If researchers are able to identify all potentially omitted variables when collecting the data, they can estimate a “rich data model,” through which they directly address endogeneity concerns and meet the “standard recommendation for limiting omitted variable bias” (Rossi 2014, p. 657). However, although these endeavors in theory may lower or rule out concerns of omitted variable bias, the success of a rich data approach depends on the researchers’ ability to identify all relevant control variables. If all potentially omitted variables cannot be identified or if such variables cannot be measured, survey researchers can employ the procedures Ebbes et al. (chapter ► [“Dealing with Endogeneity: A Nontechnical Guide for Marketing Researchers”](#)) outline.

Second, while measurement error in the dependent variable is absorbed in the residual error term, measurement error in the independent variables represents a principal form of endogeneity (Kennedy 2008; Vomberg and Wies (chapter ► [“Panel Data Analysis: A Non-technical Introduction for Marketing Researchers”](#)) in this handbook). If survey researchers rely on multiple-item measures of their independent variables, they are able to directly evaluate measurement error and through structural equation modeling can directly rule out such concerns (e.g., Grewal et al. 2013; Baumgartner and Weijters (chapter ► [“Structural Equation Modeling”](#)) in this handbook).

Measurement error may also arise in the form of biases in surveys. For instance, CMB (section [“Common Method Bias”](#)) or key informant bias (section [“Key Informant Bias”](#)) also lead to correlation between the independent variables and the residual error term. Therefore, the procedures we outlined in the previous sections can also serve as ways to account for endogeneity (e.g., procedural remedies to deal with CMB in section [“Common Method Bias”](#)).

Third, simultaneity – for instance, in the form of reverse causality – may affect cross-sectional survey data. Potential simultaneity concerns should be considered in the development stage of the questionnaire. In this regard, potential instrumental variables can be included. For instance, Homburg et al. (2012a) studied the influence of comprehensive market performance measurement systems on market knowledge. To alleviate simultaneity concerns (i.e., market knowledge drives companies’ use of

Table 3 Linking biases with steps of the survey research process: In which steps can researchers address potential biases?

Name	Definition of bias Distortion of the samples covariance structure. ...	Reasons for occurrence	Remedies	Steps in survey research process
Common method bias (section “ Common Method Bias ”)	... that arises due to the fact that the same data source was used for measuring both independent and dependent variables	Plenty of explanations, e.g.: implicit theories consistency motif satisficing	Different sources for independent and dependent variables Collect independent and dependent variables at different points in time Increase distance between independent and dependent variables Different scales for independent and dependent variables Inclusion of marker variable or direct measure of source of common method variance	Data collection Data collection Decision about question sequence Decision about the question format Decision about the survey content
Key informant bias (section “ Key Informant Bias ”)	... that arises due to the fact that the data collection has taken place through key informants providing information about a larger social unit	Positional bias Knowledge deficiencies	Adequate selection of key informants Triangulation Avoid demanding evaluations of complex social judgments (if possible in the particular research context) Report average job and company experience of participants	Data collection Decision about survey content Decision about survey content
Social desirability (section “ Social Desirability ”)	... that arises due to the tendency of respondents to reply in a manner that will be viewed favorably by others	Level of awareness: unconscious self-deception versus deliberate impression management Content domain: egoistic responde tendencies versus moralistic response tendencies	Ensuring anonymity of respondents Reinforce that there are no right or wrong answers Encourage honest answers Indirect questioning Distance between respondent and researcher With caution: Inclusion of negatively valenced items Social sensitive questions asked at the end of the questionnaire	Decision about question sequence and Decision about survey layout Decision about question wording Selection of research method Decision about question wording Decision about question sequence

Response styles (section “Response Styles”)	... that arises due to the fact that respondents, regardless of the question content, favor certain response categories	Plenty of explanations, e.g.: Confirmation bias Satisficing Impression management Item ambiguity Personality traits of respondent	<i>With</i> caution: Inclusion of balanced scales (i.e., negative and positive items) <i>Potentially</i> : Additional items for deriving measures of response styles	Decision about question wording Decision about research variables
Non-sampling bias (section “Non-sampling Bias”)	... that arises due to the fact that the population is not adequately represented in the original sample	Lack of knowledge about population	Relying on sampling frame Inclusion of variables on which selection likely appear in the questionnaire	Data collection Decision about research variables
Non-response bias (section “Nonresponse Bias”)	... that arises due to the fact that the structure of the final sample does not coincide with the structure of the original sample	Participant declines survey due to (1) personal, (2) organizational, or (3) survey factors Inability to reach participant	Increase opportunity to participate Emphasize importance of the survey Decrease perceived costs of participation Raise perceived utility of participation	Data collection Selection of survey method Cover Letter Decision about survey layout Decision about question content Decision about survey layout Pretest of the questionnaire Data collection

comprehensive market performance measurement systems), the authors relied on an instrumental variable approach and directly measured an instrumental variable in their questionnaire.

Conclusion

Surveys are flexible and powerful ways to address research questions. Indeed, in many situations it is hard to imagine how topics can be studied without directly asking participants questions. However, effective survey research requires careful survey design. As respondents usually develop answers to surveys in the course of completing a survey, investigators must craft their surveys meticulously to avoid potential biases.

To help researchers construct their surveys, we first outlined the psychology of survey response and discussed important biases. Awareness of these biases helps survey researchers develop surveys that are less susceptible to biases. We have also described the general survey process and discussed important decisions investigators face in each step, and in Table 3, we connect those insights: we delineate the types of biases and link these biases to steps in the survey process where researchers can alleviate these biasing effects.

In closing, we note that many of the issues we discussed arise in other forms of research. For example, decisions regarding question content and data collection are equally important for experimental studies (chapter ► [“Experiments in Market Research”](#) by Bornemann and Hattula; chapter ► [“Field Experiments”](#) by Valli et al.), and our discussion of CMB is applicable when experimental studies measure mediating variables (chapter ► [“Mediation Analysis in Experimental Research”](#) by Koschate-Fischer and Schwille).

Cross-References

- [Analysis of Variance](#)
- [Automated Text Analysis](#)
- [Challenges in Conducting International Market Research](#)
- [Dealing with Endogeneity: A Nontechnical Guide for Marketing Researchers](#)
- [Experiments in Market Research](#)
- [Measuring Customer Satisfaction and Customer Loyalty](#)
- [Mediation Analysis in Experimental Research](#)
- [Partial Least Squares Structural Equation Modeling](#)
- [Regression Analysis](#)
- [Structural Equation Modeling](#)

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